

Portfolio: Snap-to-grid / constrained movement

Project: <https://github.com/SP-SDU/JHotDraw.git>

Date: 2026-06-11

1. Introduction and Baseline Setup

1.1 Purpose

This portfolio follows existing JHotDraw feature Snap-to-grid / constrained movement from change request through concept location, impact analysis, refactoring, actualization, verification, and BDD testing.

Selected feature connects directly to drawing precision. User enables or uses grid constraints, then JHotDraw aligns points, rectangles, movements, and rotations through `org.jhotdraw.draw.GridConstrainer`.

1.2 Repository Setup

Item	Value
Repository path	C:\Users\User\source\repos\JHotDraw
Fork URL	https://github.com/SP-SDU/JHotDraw.git
Upstream URL	https://github.com/sweat-tek/JHotDraw.git
Feature branch	feature/snap-to-grid-portfolio
JDK version	OpenJDK 23.0.2, TODO: install/use JDK11 if strict lab environment required
Maven version	Apache Maven 3.9.9
Build command	<code>mvn clean install -DskipTests</code>
Run command	<code>mvn exec:java "-Dexec.mainClass=org.jhotdraw.samples.svg.Main" in jhotdraw-samples/jhotdraw-samples-misc</code>

Required build command:

```
mvn clean install -DskipTests
```

Output summary:

```
[INFO] Reactor Summary for jhotdraw 9.1-SNAPSHOT:
[INFO] jhotdraw ..... SUCCESS
[INFO] jhotdraw-api ..... SUCCESS
[INFO] jhotdraw-utils ..... SUCCESS
[INFO] jhotdraw-xml ..... SUCCESS
[INFO] jhotdraw-datatransfer ..... SUCCESS
[INFO] jhotdraw-actions ..... SUCCESS
[INFO] jhotdraw-core ..... SUCCESS
[INFO] jhotdraw-gui ..... SUCCESS
[INFO] jhotdraw-app ..... SUCCESS
[INFO] jhotdraw-samples ..... SUCCESS
[INFO] jhotdraw-samples-misc ..... SUCCESS
[INFO] jhotdraw-samples-mini ..... SUCCESS
[INFO] BUILD SUCCESS
```

Required run command:

```
cd jhotdraw-samples/jhotdraw-samples-misc
mvn exec:java "-Dexec.mainClass=org.jhotdraw.samples.svg.Main"
```

Output proof:

```
[INFO] --- exec:3.1.0:java (default-cli) @ jhotdraw-samples-misc ---
Warning ResourceBundleUtil[org.jhotdraw.app.Labels].getIconProperty "application.about.icon" not found.
Warning ResourceBundleUtil[org.jhotdraw.draw.Labels].getIconProperty "edit.moveConstrainedEast.icon" not found.
Warning ResourceBundleUtil[org.jhotdraw.draw.Labels].getIconProperty "edit.moveConstrainedWest.icon" not found.
Warning ResourceBundleUtil[org.jhotdraw.draw.Labels].getIconProperty "edit.moveConstrainedNorth.icon" not found.
Warning ResourceBundleUtil[org.jhotdraw.draw.Labels].getIconProperty "edit.moveConstrainedSouth.icon" not found.
Process exceeded 20000 ms timeout because Swing GUI stays open.
```

1.3 Baseline Result

Build succeeded. SVG application reached Swing action/resource initialization and stayed open until timeout, which is expected for GUI process.

Lab review: IntroLab met with build, run command, branch, Maven/JDK evidence. KISS kept setup changes separate from feature logic.

2. Change Request

2.1 Requested Feature

Requested feature:

- Snap-to-grid / constrained movement

2.2 Contained Subfeatures

Contained subfeatures:

- Point snapping to nearest grid intersection.
- Rectangle placement by nearest edge to grid.
- Directional movement by constrained grid unit.
- Rotation angle snapping by configured theta.
- Visible grid toggle and grid-size UI in SVG sample.

2.3 User Story

As a drawing user, I want snap-to-grid and constrained movement, so that I can place figures with consistent alignment.

2.4 Details / Functional Requirements

Requirement	Behavior
User can toggle grid	<code>ToggleGridAction.actionPerformed</code> calls <code>DrawingView.setConstrainerVisible</code> .
User can set grid size	<code>ViewToolBar</code> updates <code>GridConstrainer.setWidth</code> and <code>setHeight</code> .
Selected objects affected	Points, figure bounds, rectangles, and rotation operations routed through <code>Constrainer</code> .
Expected result	Coordinates align to configured grid size or theta.
Edge case	<code>rotateAngle</code> rejects null <code>RotationDirection</code> with <code>IllegalArgumentException</code> .
Edge case	Directional point translation east changes x coordinate but preserves y coordinate.

2.5 Acceptance Criteria

- [x] Given point 14,13 and grid 10 by 5, when `constrainPoint` runs, then point becomes 10,15.
- [x] Given rectangle 13,26,12,8 and grid 10 by 10, when `constrainRectangle` runs, then rectangle moves to 10,22 without resizing.
- [x] Given point 13,26 and grid 10 by 10, when `translatePoint EAST` runs, then point becomes 20,26.
- [x] Given angle 0.70 and theta $\pi/4$, when `constrainAngle` runs, then angle becomes $\pi/4$.
- [x] Given null rotation direction, when `rotateAngle` runs, then `IllegalArgumentException` is captured.

Lab review: ChangeReqLab met with selected existing JHotDraw feature, user story, requirements, and acceptance criteria. KISS scope excludes new user-facing behavior.

3. Continuous Integration

3.1 CI Purpose

CI fits this repository because JHotDraw is a Maven multi-module project. Pull requests should build all modules and run tests automatically so integration errors appear before merge.

3.2 GitHub Actions Workflow

File: `.github/workflows/maven.yml`

```

name: Maven CI

on:
  pull_request:
  push:
    branches:
      - develop
      - main

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository
        uses: actions/checkout@v4

      - name: Set up JDK 11
        uses: actions/setup-java@v4
        with:
          distribution: temurin
          java-version: '11'
          cache: maven

      - name: Build with Maven
        run: mvn -B clean install -DskipTests

      - name: Run tests
        run: mvn -B test

```

3.3 CI Result

Item	Value
Trigger	Pull request and pushes to develop or main.
Build command	mvn -B clean install -DskipTests
Test command	mvn -B test
Local equivalent result	mvn test succeeded with all modules green.
Remote workflow result	TODO: push branch and record GitHub Actions URL after review.

Lab review: CILab met by adding Maven GitHub Actions workflow and local verification commands. KISS uses one workflow and no repository secrets because current dependency resolution works without `.maven-settings.xml` locally.

4. Concept Location

4.1 Method

Concept location used static search and runtime launch of the SVG sample. Search terms included `GridConstrainer`, `getConstrainer`, `setConstrainerVisible`, `ToggleGridAction`, and `MoveConstrainedAction`. Runtime action is SVG app startup with grid-related actions loaded.

TODO: Add IDE debugger screenshot or stack trace if instructor requires dynamic debugger artifact.

4.2 Triggering User Action

Feature can be triggered by selecting drawing objects and using constrained movement actions such as `edit.moveConstrainedEast`, or by toggling grid visibility through SVG toolbar button created by `ButtonFactory.createToggleGridButton(view)`.

4.3 Initial Concept Classes

Domain Class	Package	How Found	Responsibility
<code>GridConstrainer</code>	<code>org.jhotdraw.draw</code>	Search, call hierarchy, tests	Implements grid snap and rotation constraints.
<code>Constrainer</code>	<code>org.jhotdraw.draw</code>	<code>DrawingView.getConstrainer</code> contract	Strategy interface for editing constraints.
<code>DrawingView</code>	<code>org.jhotdraw.draw</code>	Grid visibility and constrainer accessors	View-level owner of current constrainer.
<code>DefaultDrawingView</code>	<code>org.jhotdraw.draw</code>	Implementation of <code>DrawingView</code>	Returns visible or invisible constrainer.
<code>ToggleGridAction</code>	<code>org.jhotdraw.draw.action</code>	Grid action search	Toggles visible constrainer.
<code>MoveConstrainedAction</code>	<code>org.jhotdraw.draw.action</code>	Constrained movement search	Moves selection by constrained unit.
<code>AbstractTool</code>	<code>org.jhotdraw.draw.tool</code>	<code>constrainPoint</code> call	Delegates mouse coordinates to current constrainer.
<code>ViewToolBar</code>	<code>org.jhotdraw.samples.svg.gui</code>	SVG toolbar path	Exposes grid UI controls.
<code>SVGApplicationModel</code>	<code>org.jhotdraw.samples.svg</code>	SVG startup path	Creates default <code>GridConstrainer(12, 12)</code> .

4.4 Concept Location Summary

Entry point for runtime GUI is `org.jhotdraw.samples.svg.Main`, which uses SVG application model and view UI. Domain center is `GridConstrainer`. UI classes are `ViewToolBar` and `ToggleGridAction`. Model/data classes are `Point2D.Double`, `Rectangle2D.Double`, and drawing figures reached through `DrawingView` and tools.

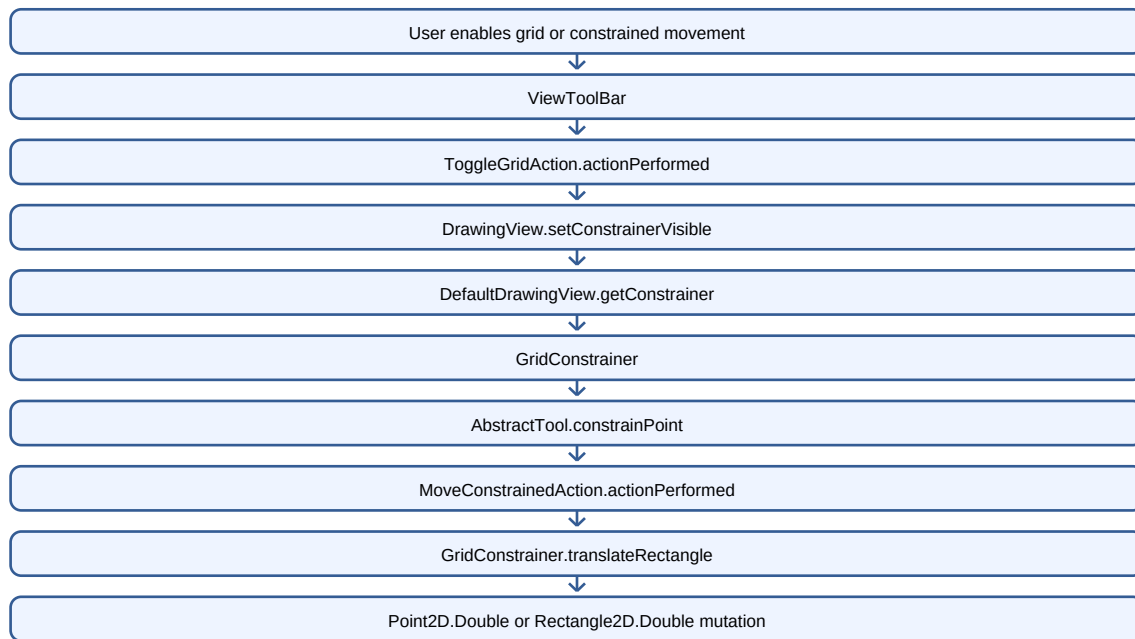
Classes excluded: figure subclasses such as `RectangleFigure` and `EllipseFigure`, because snapping mutates geometric coordinates through constrainer logic and does not depend on concrete figure type.

Lab review: CLLab met by listing exact concept classes and responsibilities. KISS keeps concept set limited to classes that participate in selected feature.

5. Feature Call Tree / Interaction View

5.1 Call Tree

Rendered Mermaid diagram



5.2 Important Calls

Step	Caller	Method	Callee	Purpose
1	ViewToolBar	createDisclosedComponent	ButtonFactory.createToggleGridButton	Creates toolbar UI for grid.
2	ToggleGridAction	actionPerformed	DrawingView.setConstrainerVisible	Turns visible grid constrainer on or off.
3	DefaultDrawingView	getConstrainer	GridConstrainer	Resolves current constrainer.
4	AbstractTool	constrainPoint	Constrainer.constrainPoint	Applies grid snapping during mouse editing.
5	MoveConstrainedAction	actionPerformed	Constrainer.translateRectangle	Applies constrained keyboard movement.
6	GridConstrainer	constrainPoint	Point2D.Double	Snaps point coordinates.
7	GridConstrainer	constrainRectangle	Rectangle2D.Double	Aligns rectangle edge to grid.

5.3 Scope Count

Feature call tree spans 5 packages, 8 classes/interfaces, and 10 relevant methods.

5.4 Interpretation

Central classes are `GridConstrainer`, `Constrainer`, and `DrawingView`. Incidental classes are SVG toolbar and actions because they trigger or expose feature but do not define snapping math.

Lab review: Call tree met using small static/runtime path. KISS avoids tracing unrelated figure rendering and persistence classes.

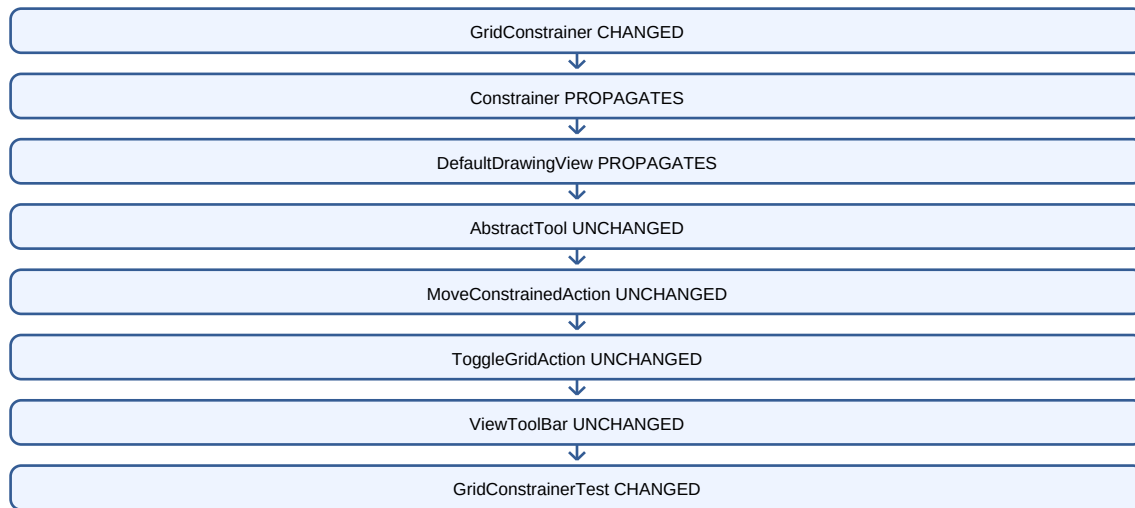
6. Impact Analysis

6.1 Impact Analysis Method

Starting point was GridConstrainer. Neighbors were classified using CHANGED, PROPAGATES, UNCHANGED, and NEXT. Because public method signatures stayed unchanged, only implementation and tests needed modification.

6.2 Impact Diagram

Rendered Mermaid diagram



6.3 Estimated Impact Set

Class	Package	Mark	Reason
GridConstrainer	org.jhotdraw.draw	CHANGED	Extracted coordinate snapping helper.
GridConstrainerTest	org.jhotdraw.draw	CHANGED	Added unit and BDD coverage.
Constrainer	org.jhotdraw.draw	PROPAGATES	Public contract unchanged.
DrawingView	org.jhotdraw.draw	PROPAGATES	Exposes current constrainer.
DefaultDrawingView	org.jhotdraw.draw	PROPAGATES	Holds and returns GridConstrainer.
MoveConstrainedAction	org.jhotdraw.draw.action	UNCHANGED	Uses existing API.
ToggleGridAction	org.jhotdraw.draw.action	UNCHANGED	Toggle behavior unchanged.
AbstractTool	org.jhotdraw.draw.tool	UNCHANGED	Delegation unchanged.
ViewToolBar	org.jhotdraw.samples.svg.gui	UNCHANGED	UI contract unchanged.
SVGApplicationModel	org.jhotdraw.samples.svg	UNCHANGED	Constructor use unchanged.

6.4 Package Visit Table

Package name	# classes visited	Comments
org.jhotdraw.draw	5	Core domain contract and GridConstrainer implementation.
org.jhotdraw.draw.action	2	Trigger paths for grid toggle and constrained movement.
org.jhotdraw.draw.tool	1	Pointer editing path.
org.jhotdraw.samples.svg	1	SVG app model creates default grid.
org.jhotdraw.samples.svg.gui	1	SVG toolbar exposes feature controls.

6.5 Impact Summary

Minimal production change set is one class: GridConstrainer. Risk area is preserving exact rounding behavior in constrainPoint, constrainRectangle, translatePoint, and constrainAngle. Classes intentionally left unchanged are UI actions, view implementation, and tool delegation classes.

Lab review: AnalysisLab met with package visit table and estimated impact set. KISS change surface stayed small.

7. Refactoring

7.1 Refactoring Goal

Improve readability of grid snapping calculation without changing external behavior.

7.2 Code Smell

Smell	Location	Evidence	Consequence
Duplicated expression	GridConstrainer.constrainPoint	Math.round(p.x / width) * width and Math.round(p.y / height) * height repeat same rule.	Future coordinate snapping edits can drift between axes.

7.3 Refactoring Strategy

Plan was to extract private helper method, call it for x and y, then run focused and full tests. Public API, constructors, property events, and UI behavior stayed unchanged.

7.4 Refactoring Pattern Applied

Refactoring	Location	Purpose
Extract Method	GridConstrainer.constrainPoint	Name and isolate coordinate snap formula.

7.5 Before Snippet

```
@Override
public Point2D.Double constrainPoint(Point2D.Double p, Figure... figure) {
    p.x = Math.round(p.x / width) * width;
    p.y = Math.round(p.y / height) * height;
    return p;
}
```

7.6 After Snippet

```
@Override
public Point2D.Double constrainPoint(Point2D.Double p, Figure... figure) {
    p.x = constrainCoordinate(p.x, width);
    p.y = constrainCoordinate(p.y, height);
    return p;
}

private double constrainCoordinate(double value, double gridSize) {
    return Math.round(value / gridSize) * gridSize;
}
```

7.7 Refactoring Result

Improved code names the snap formula and removes duplicate math. External behavior stayed same because helper contains exact original expression. `mvn test` passed.

Lab review: RefactoringLab met with smell, pattern, strategy, before/after, and tests. KISS changed only six production lines.

8. Actualization

8.1 Implementation Summary

Actualization incorporated refactoring and tests into existing code. Feature behavior was not expanded. Test and CI support were added around existing snap-to-grid behavior.

8.2 Change Propagation

Primary Change	Propagated Change	Reason
Extracted constrainCoordinate	No caller changes	Helper is private and preserves public API.
Added JUnit 4/JGiven/AssertJ dependencies	Parent Surefire argLine added	JGiven needs Java module opening under Java 23.
Added GridConstrainerTest	CI now runs mvn test	Automated verification for selected feature.

8.3 SOLID Principles in Case Study

Principle	JHotDraw Example	Explanation
SRP	GridConstrainer	Owns grid constraint math and grid drawing only.
OCP	Constrainer	New constraint strategies can implement interface without changing tools.
LSP	GridConstrainer implements Constrainer	Tools use any Constrainer through interface.
ISP	Constrainer	Editing code depends only on constraint operations it needs.
DIP	DrawingView depends on Constrainer	View and tools depend on abstraction, not only GridConstrainer.

8.4 Clean Architecture in JHotDraw Context

Layer / Concern	JHotDraw Example	Responsibility
UI	ViewToolBar, buttons, grid size field	User interaction.
Controller / Action	ToggleGridAction, MoveConstrainedAction	Command handling.
Domain Model	DrawingView, Constrainer, GridConstrainer	Drawing editing state and constraints.
Infrastructure	Maven, GitHub Actions, resource bundles	Build, CI, resource loading.

8.5 Clean Code Principles

Visible clean code principles: meaningful names, small helper method, low duplication, clear responsibility, and readable BDD test names.

Lab review: ActualizationLab met with SOLID, clean architecture, and propagation. KISS actualization avoids feature creep.

9. Verification / Unit Testing

9.1 Verification Goal

Refactoring should preserve observed behavior. Same feature behavior should pass before and after refactoring.

9.2 Classes Under Test

Changed classes:

- org.jhotdraw.draw.GridConstrainer

Test classes:

- org.jhotdraw.draw.GridConstrainerTest

9.3 Test Strategy

Test strategy uses JUnit 4 plus JGiven for BDD-formatted steps and AssertJ for readable assertions. Tests cover best-case snapping, rectangle boundary behavior, directional translation, theta rounding, and invalid rotation direction.

9.4 Test Case Table

Test Class	Test Method	Type	Purpose	Expected Result
GridConstrainerTest	givenPointNearGrid_whenConstrainPoint_thenSnapsToNearestGrid	best-case	Point snap math	14, 13 becomes 10, 15.
GridConstrainerTest	givenRectangleNearGrid_whenConstrainRectangle_thenMovesNearestEdgeToGrid	boundary	Edge chosen by nearest movement	Rectangle moves to 10, 22, same size.
GridConstrainerTest	givenDirectionEast_whenTranslatePoint_thenMovesOneGridCellEast	boundary	Directional movement	X becomes 20, Y remains 26.
GridConstrainerTest	givenTheta_whenConstrainAngle_thenRoundsToNearestStep	best-case	Rotation snapping	0.70 becomes pi/4.
GridConstrainerTest	givenNullRotationDirection_whenRotateAngle_thenThrowsIllegalArgumentException	boundary	Invalid input	IllegalArgumentException.

9.5 Important Test Snippet

```
@Test
public void givenPointNearGrid_whenConstrainPoint_thenSnapsToNearestGrid() {
```

```

        given().a_grid_constraint(10, 5)
            .and().a_point(14, 13);

        when().the_point_is_constrained();

        then().the_point_is(10, 15);
    }

```

This test proves selected feature snaps a drawing point to configured grid dimensions.

9.6 Test Result

```

mvn test

[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0 -- org.jhotdraw.geom.BezierPathNGTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0 -- jhotdraw-core TestSuite
[INFO] Reactor Summary for jhotdraw 9.1-SNAPSHOT:
[INFO] jhotdraw ..... SUCCESS
[INFO] jhotdraw-api ..... SUCCESS
[INFO] jhotdraw-utils ..... SUCCESS
[INFO] jhotdraw-xml ..... SUCCESS
[INFO] jhotdraw-datatransfer ..... SUCCESS
[INFO] jhotdraw-actions ..... SUCCESS
[INFO] jhotdraw-core ..... SUCCESS
[INFO] jhotdraw-gui ..... SUCCESS
[INFO] jhotdraw-app ..... SUCCESS
[INFO] jhotdraw-samples ..... SUCCESS
[INFO] jhotdraw-samples-misc ..... SUCCESS
[INFO] jhotdraw-samples-mini ..... SUCCESS
[INFO] BUILD SUCCESS

```

9.7 Verification Summary

Tests prove the exact coordinate and angle behavior of GridConstrainer remains correct after extracting the helper. Full Maven tests prove integration across modules remains green.

Lab review: TestLab1 met with class-level tests, best-case and boundary cases, assertions, and command evidence. KISS uses one focused test class.

10. BDD Scenarios

10.1 User Story Mapping

User Story	Given	When	Then
As a drawing user, I want snap-to-grid and constrained movement, so that I can place figures with consistent alignment.	Grid constrainer and off-grid point	Point is constrained	Point moves to grid intersection
Same story	Grid constrainer and off-grid rectangle	Rectangle is constrained	Nearest edge moves to grid
Same story	Grid constrainer and point	Point moves east	X moves to next grid cell

10.2 Scenario 1

```

Scenario: Point snaps to nearest grid intersection
Given a grid constrainer with width 10 and height 5
And a point at 14, 13
When the point is constrained
Then the point is moved to 10, 15

```

10.3 Scenario 2

```

Scenario: Rectangle moves nearest edge to grid
Given a grid constrainer with width 10 and height 10
And a rectangle at 13, 26 with size 12 by 8
When the rectangle is constrained
Then the rectangle is moved to 10, 22 without changing size

```

10.4 JGiven Automation

```

@Test
public void givenRectangleNearGrid_whenConstrainRectangle_thenMovesNearestEdgeToGrid() {
    given().a_grid_constraint(10, 10)
        .and().a_rectangle(13, 26, 12, 8);

    when().the_rectangle_is_constrained();

    then().the_rectangle_is(10, 22, 12, 8);
}

```


10.5 Stage Explanation

Stage	Method	Purpose
Given	<code>a_grid_constraint</code>	Creates feature object with width, height, and theta.
Given	<code>a_point, a_rectangle</code>	Creates drawing geometry state.
When	<code>the_point_is_constrained</code>	Runs selected feature operation.
When	<code>the_rectangle_is_constrained</code>	Runs rectangle snapping operation.
Then	<code>the_point_is, the_rectangle_is</code>	Verifies exact snapped coordinates.

10.6 BDD Summary

BDD scenarios map directly to user story outcome: aligned drawing geometry. They are automated through `GridConstrainerTest`, not only documented as text.

Lab review: BDDLab met with Given-When-Then scenarios, JGiven automation, and AssertJ assertions. KISS uses BDD at domain level instead of brittle Swing UI automation.

11. Final Exam-Oriented Summary

11.1 Feature Summary

Snap-to-grid / constrained movement aligns drawing coordinates to configured grid units. SVG app creates and exposes `GridConstrainer` through toolbar and view actions. Tools and movement actions depend on `Constrainer` abstraction, so snapping behavior stays centralized. Refactoring extracted coordinate snapping math without changing public behavior.

11.2 Main Classes

Class	Role
<code>GridConstrainer</code>	Core snap-to-grid implementation.
<code>Constrainer</code>	Strategy interface used by tools and views.
<code>DrawingView</code>	Exposes current constrainer.
<code>DefaultDrawingView</code>	Stores visible/invisible constrainers.
<code>ToggleGridAction</code>	Toggles grid visibility.
<code>MoveConstrainedAction</code>	Moves selected figures by constrained units.
<code>AbstractTool</code>	Delegates pointer coordinates to constrainer.
<code>ViewToolBar</code>	SVG UI for grid toggle and size.

11.3 Main Change

Main production change is Extract Method in `GridConstrainer.constrainPoint`: duplicated rounding formula moved into private `constrainCoordinate` helper.

11.4 Why Behavior Is Preserved

Helper uses exact original expression `Math.round(value / gridSize) * gridSize`. Tests cover point, rectangle, directional movement, angle, and invalid input behavior. Full `mvn test` passed.

11.5 What To Remember For Exam

- Selected feature: Snap-to-grid / constrained movement.
- Entry point class: `GridConstrainer` for domain behavior, `ViewToolBar/ToggleGridAction` for UI path.
- Domain classes: `Constrainer`, `DrawingView`, `DefaultDrawingView`, `GridConstrainer`.
- Impact set: one production class changed, one test class added, CI/build metadata changed.
- Refactoring used: Extract Method.
- Tests proving behavior: `GridConstrainerTest` plus `mvn test`.
- BDD scenario: Given off-grid geometry, When constrained, Then coordinates align to grid.

12. Appendix

12.1 Full CI File

```
name: Maven CI

on:
  pull_request:
  push:
    branches:
      - develop
      - main

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository
        uses: actions/checkout@v4

      - name: Set up JDK 11
        uses: actions/setup-java@v4
        with:
          distribution: temurin
          java-version: '11'
          cache: maven

      - name: Build with Maven
        run: mvn -B clean install -DskipTests

      - name: Run tests
        run: mvn -B test
```

12.2 Full Test Output

See evidence/test-results.txt.

12.3 Extra Diagrams

See evidence/call-tree.md and evidence/impact-analysis-table.md.

12.4 Links / Commits

Artifact	Link
Repository	https://github.com/SP-SDU/JHotDraw.git
Branch	TODO: push branch and add branch URL
Pull request	TODO: create PR after review
Commit before refactor	TODO: commit after review and record hash
Commit after refactor	TODO: commit after review and record hash

12.5 Final Quality Checklist

- [x] Every lab portfolio requirement has section.
- [x] All known class names are exact.
- [x] All known package names are exact.
- [x] Code snippets are readable.
- [x] No screenshot used as primary evidence.
- [x] Call tree included.
- [x] Concept location table included.
- [x] Impact analysis package table included.
- [x] Refactoring smell and pattern explained.
- [x] SOLID and Clean Architecture tied to JHotDraw.
- [x] Unit tests documented.
- [x] BDD Given-When-Then scenarios documented.
- [x] PDF generated from Markdown after content completion.

- [] TODO: Remote CI URL captured after push.